Sampling and assignment

A population is the group about which a study wants to draw conclusions. A sample is the group actually observed. Random sampling is a method of selecting observations from a population using a random mechanism with defined selection probabilities. It can support population inference, subject to the sampling design, nonresponse, and other limitations.

Random assignment is a method of assigning study participants to treatment conditions using a random mechanism. It aims to make treatment groups comparable in expectation before treatment. It does not guarantee identical groups in every realized experiment.

Random sampling and random assignment answer different questions. Sampling concerns who enters the study; assignment concerns which treatment participants receive. A study can use either, both, or neither.

Example: Students volunteer for a study of a study-planning app. The researchers randomly assign the volunteers to use the app or to continue their usual routine. This is random assignment within a volunteer sample. The design can support a causal comparison for the study participants under appropriate assumptions, but the volunteers may differ from the wider student population.

Example: Researchers randomly sample students from a university roster and observe their existing study habits. This is random sampling without random assignment of habits. A representative sample does not by itself turn an observed relationship between habits and grades into a causal effect.

Questions to explain

Which process affects group comparability? Which process affects population coverage? Can a randomized experiment still have limited generalizability?

Association, confounding, and limits

An association means two variables vary together in the observed data. A causal claim says that changing one variable would change another under specified conditions. An association alone does not establish such a claim.

A confounder is a variable related to the treatment or exposure and the outcome in a way that can distort the observed association. In a study of app use and grades, prior motivation may influence both the decision to use the app and academic performance.

Random assignment addresses systematic pre-treatment differences between groups in expectation. It does not automatically solve every problem: differential dropout, noncompliance, measurement problems, or interference between participants can still complicate interpretation.

Suppose volunteers randomly assigned to the app group have higher average grades at the end of a well-conducted study. This is stronger evidence for a causal effect in that study than a simple observation that app users have higher grades. Extending the result to all college students requires considering the population, sample, implementation, and setting.

A careful conclusion distinguishes the design's causal support from the scope of generalization. “Random assignment” does not mean “random sample,” and “random sample” does not mean “causation established.”

Questions to explain

How could motivation create an association without an app effect? Why is random assignment useful even though realized groups may differ by chance? What additional question remains when generalizing beyond volunteers?